EMNLP 2024

Do LLMs suffer from Multi-Party Hangover?

A Diagnostic Approach to Addressee Recognition and Response Selection in Conversations

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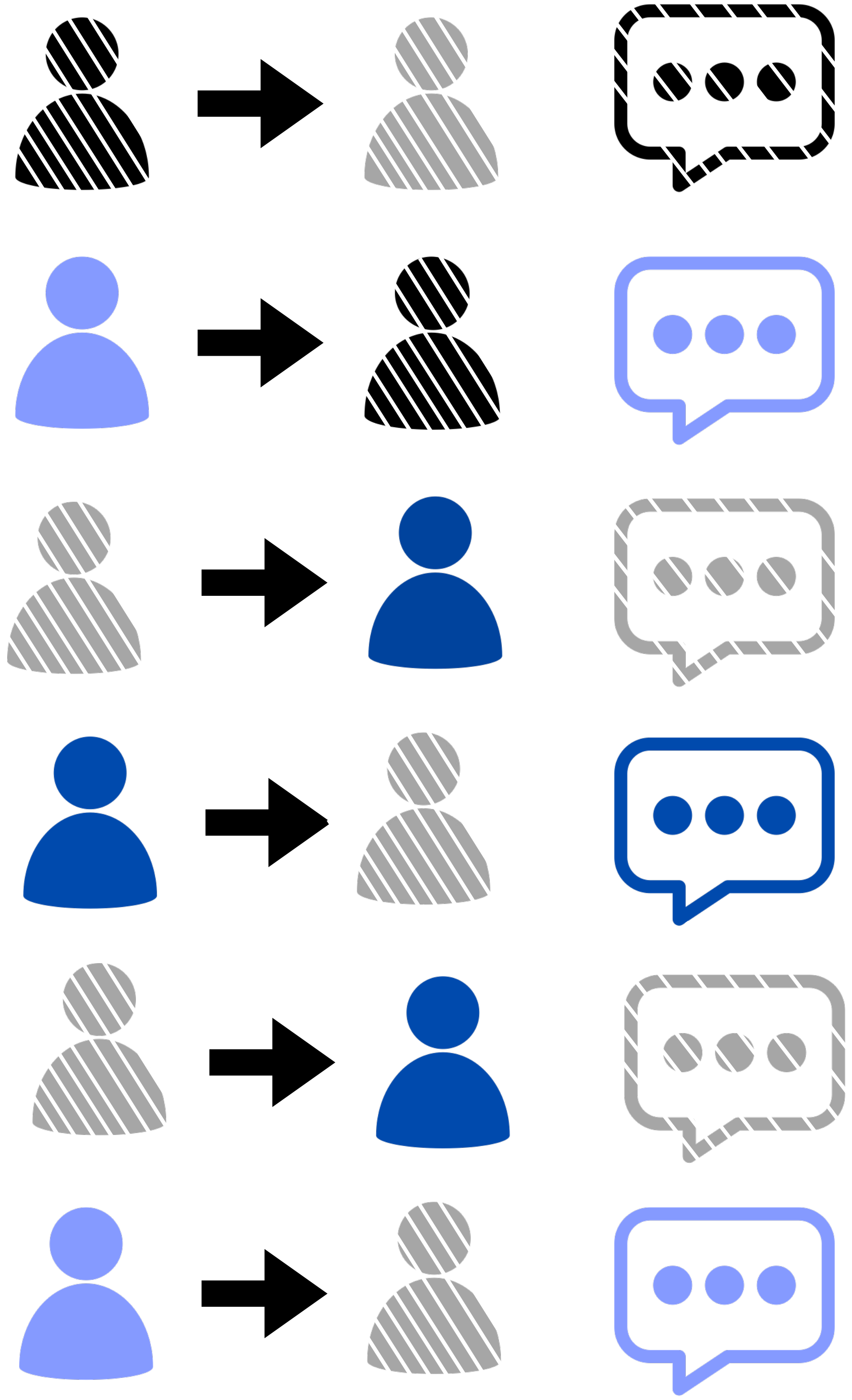
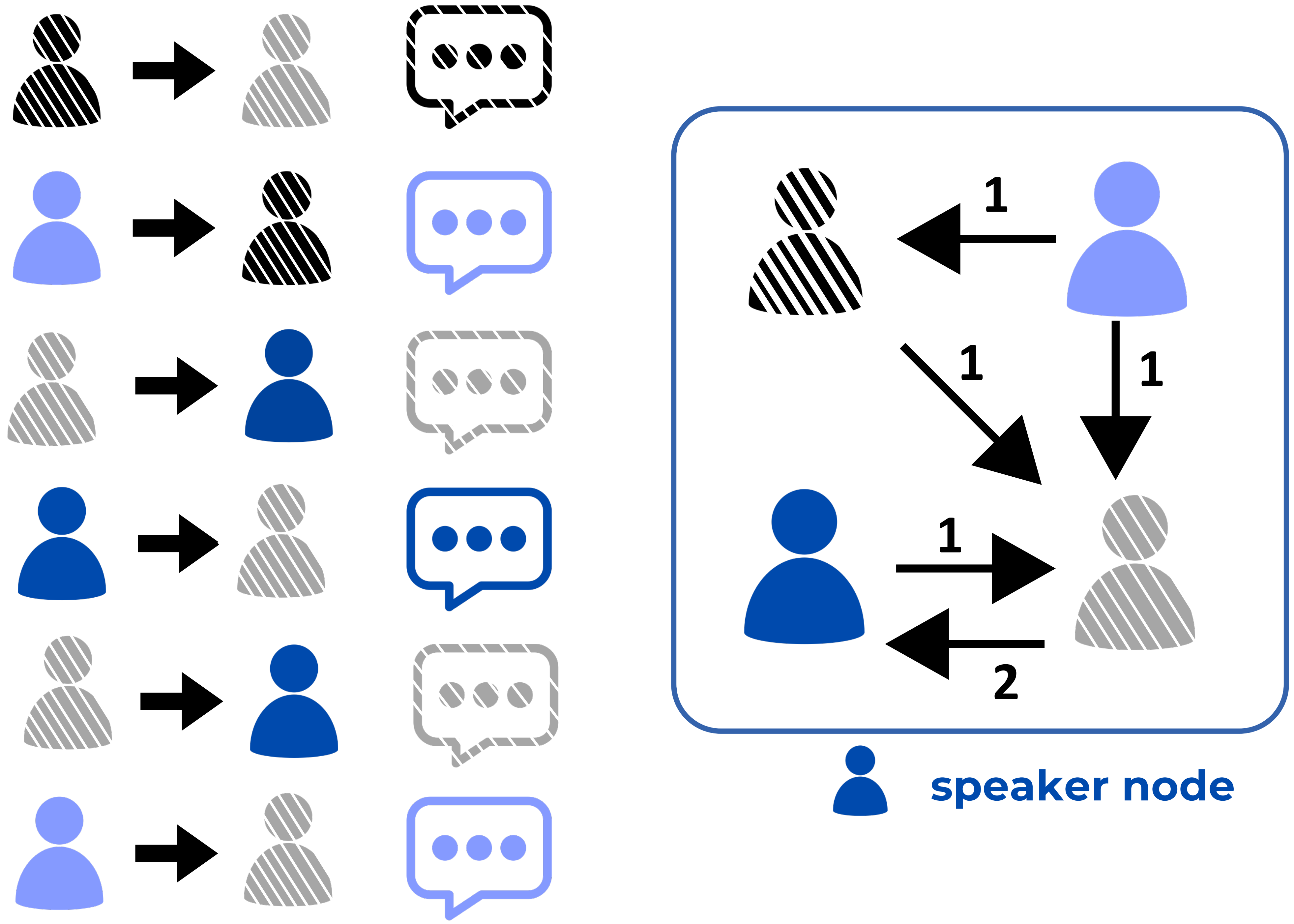
Multi-Party Conversation



Multi-Party conversations:

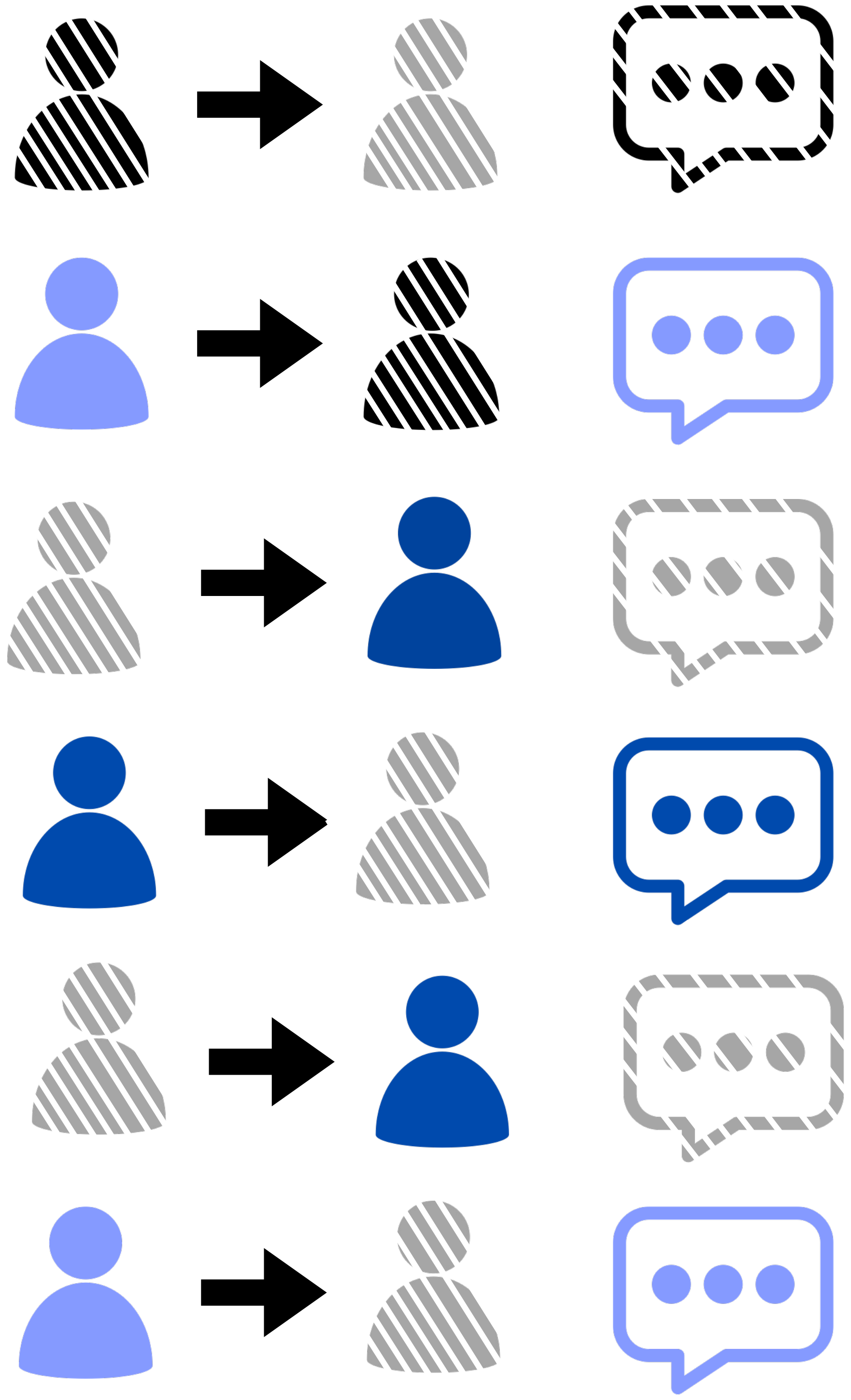
- are multi-turn
- have multiple speakers

Interaction Graph

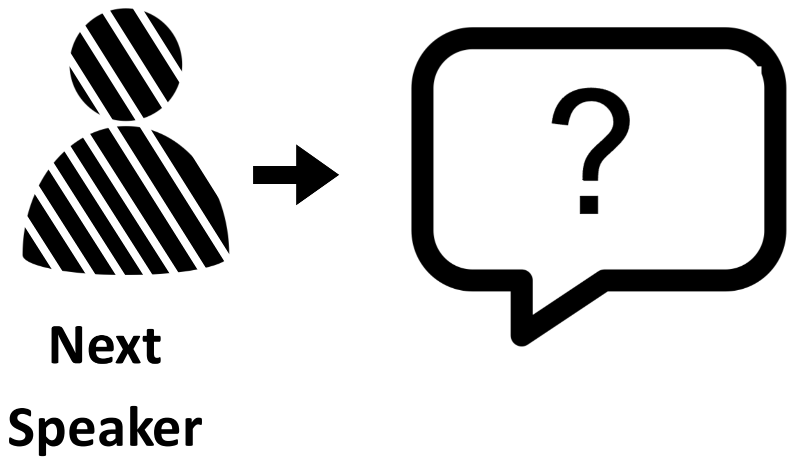


Capture the ongoing user dynamics

Task 1: Response Selection

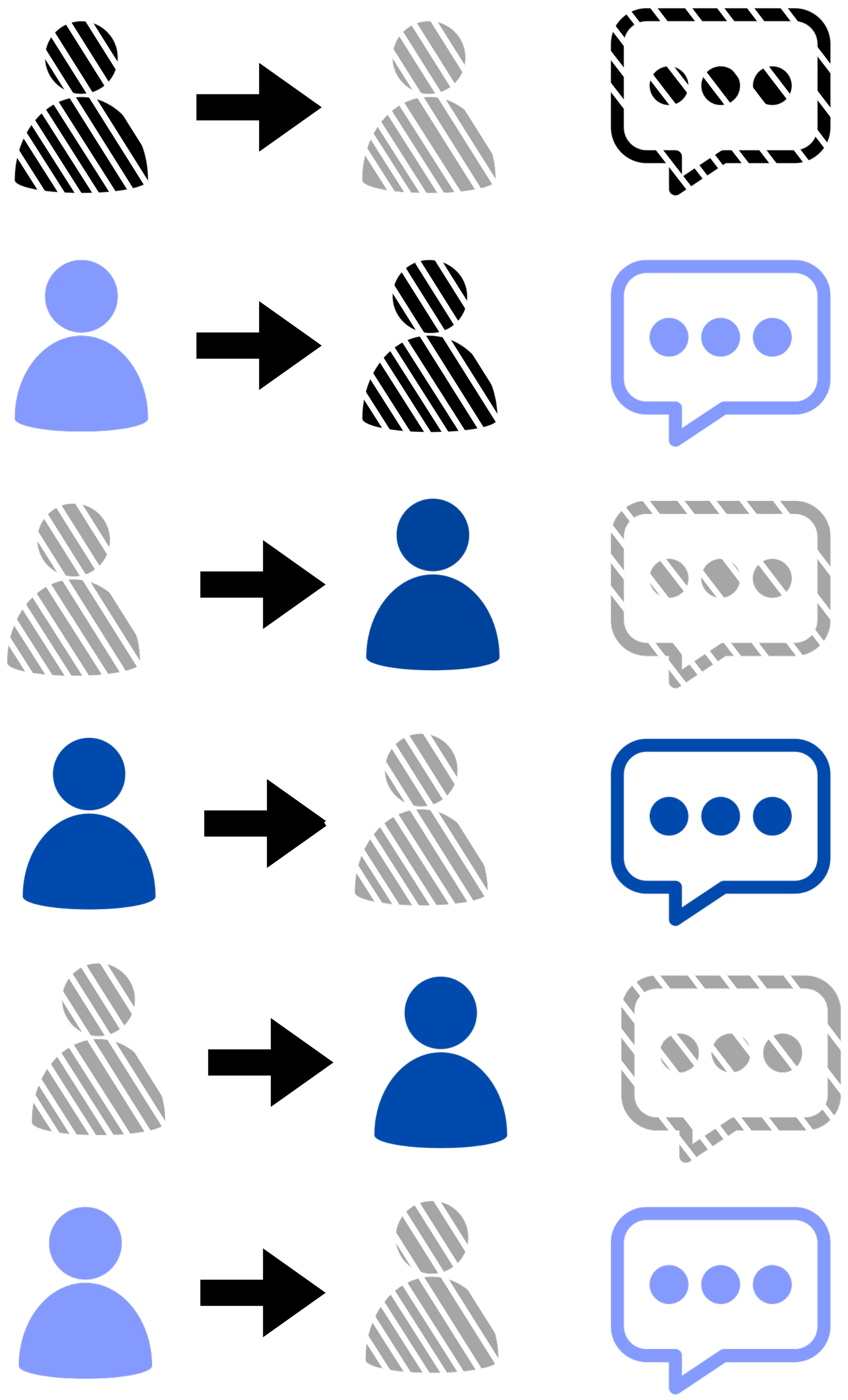


Given 2 candidate responses, choose the correct one

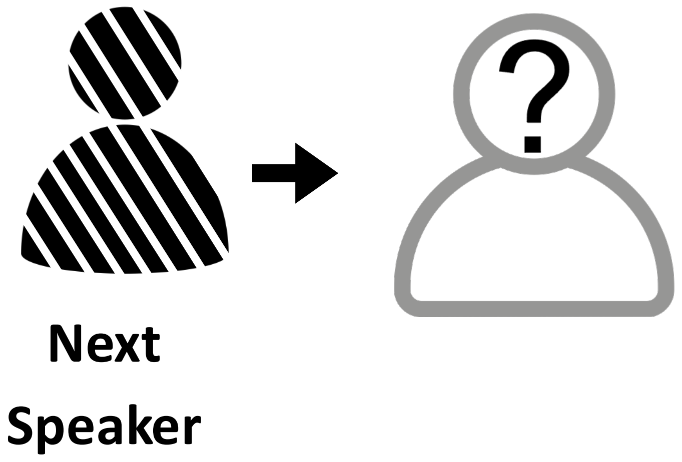


Linguistic perspective

Task 2: Addressee Recognition

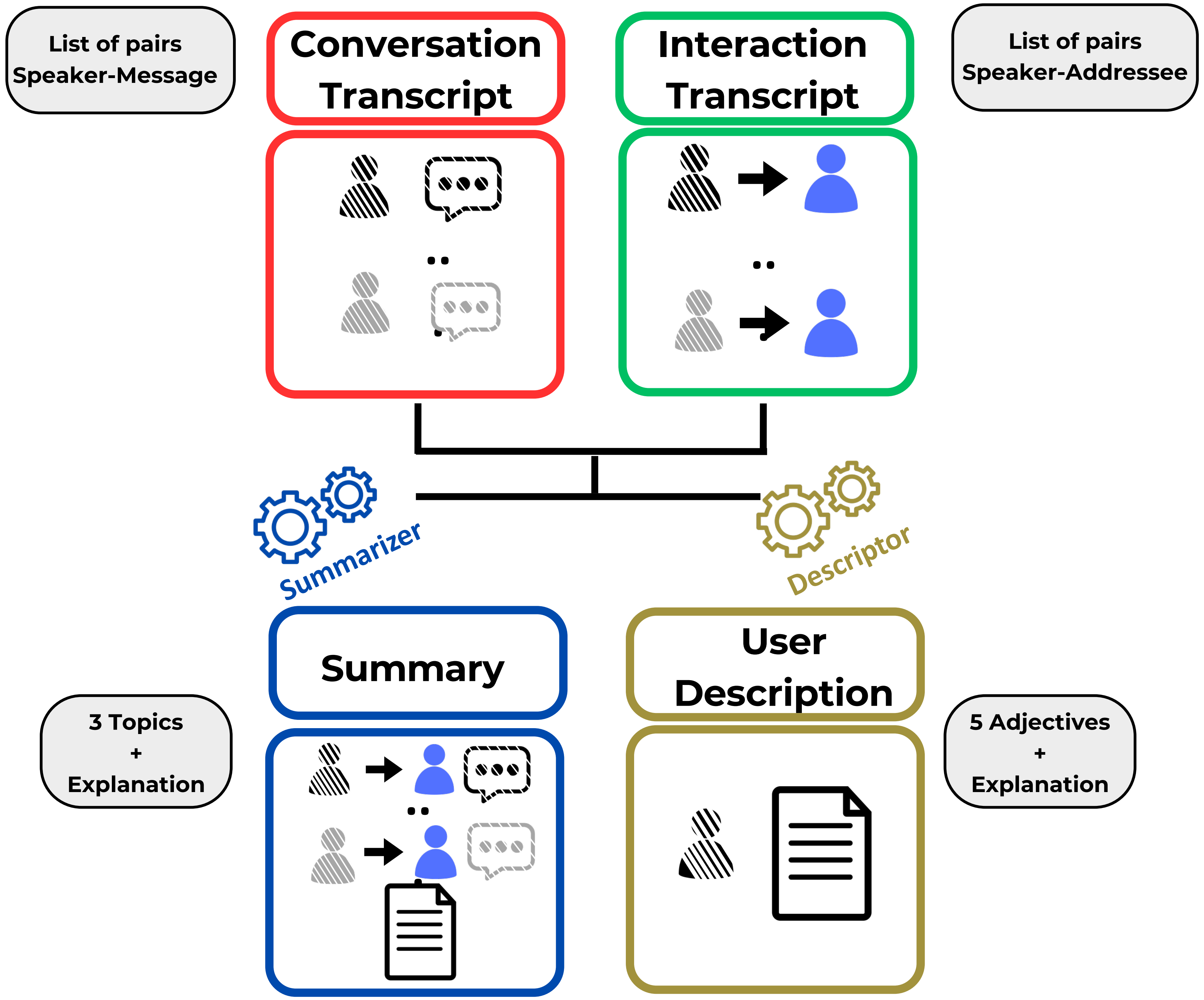


Given *n* candidate addressees, choose the correct one



Structural perspective

Input Information



What happens if we don't have access to the original messages?

- **data minimization policy**
- **privacy preserving**

We used a prompted LLM to generate Summary and User Description.

Research Questions

RQ1

How do LLMs perform in classification tasks involving MPCs in a zero-shot setting, using different input combinations to capture textual and structural information?

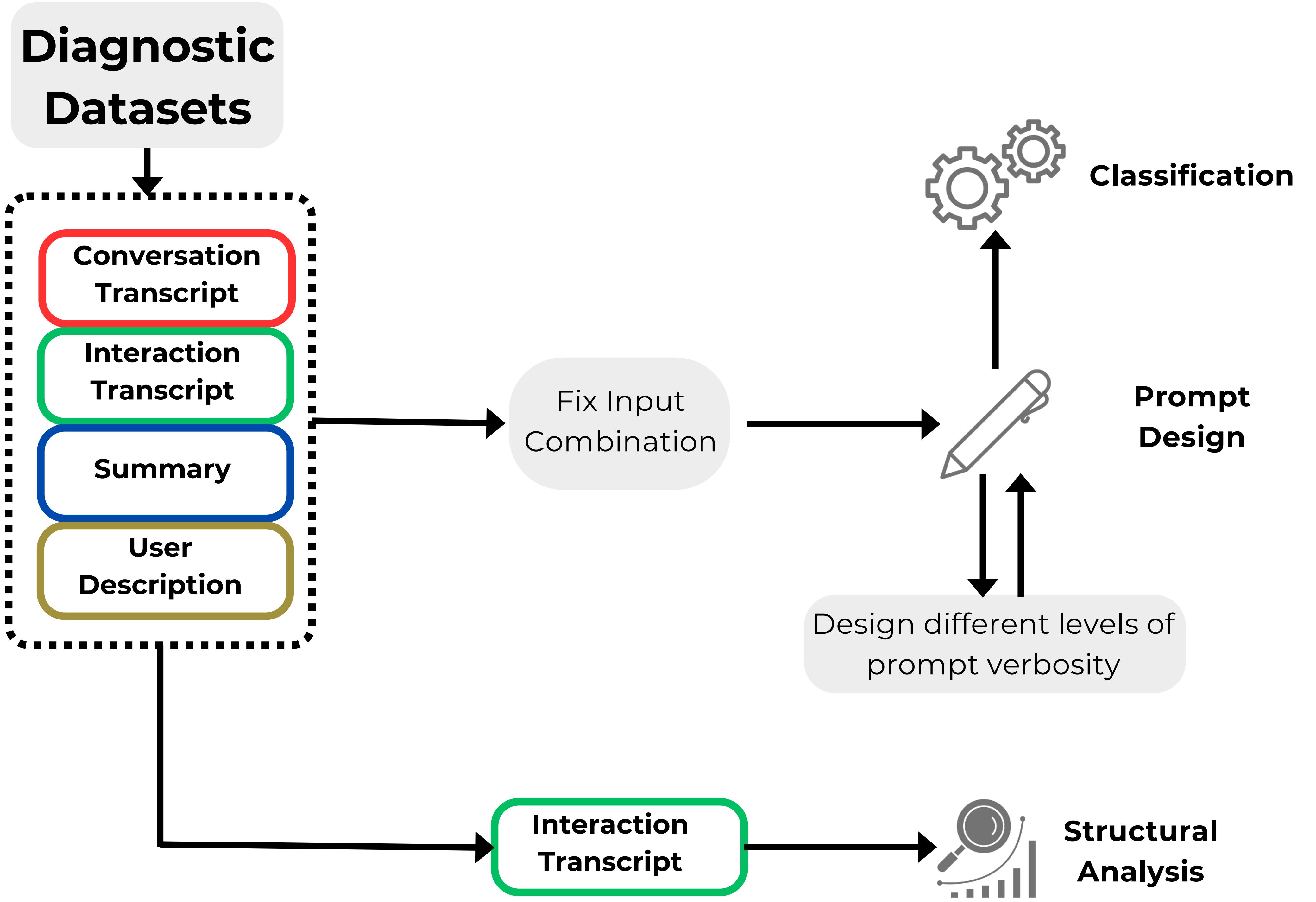
RQ2

What is the model sensitivity to different prompt formulations when classifying MPCs?

RQ3

How does structural complexity of the conversation affect classification performance?

Pipeline



Llama2-13b-chat

Diagnostic Datasets

Starting from a big MPC corpus from **Ubuntu Internet Relay Chat** (Ouchi and Tsuboi, 2016) we retrieved 4 different diagnostic datasets:

- Each with a **fixed number of user** for each conversation (Ubuntu 3/4/5/6), respectively with 1200, 635, 520 and 350 conversations.
- **Fixed number of messages** (15 each).
- Good **structural variety.**

Input Combinations

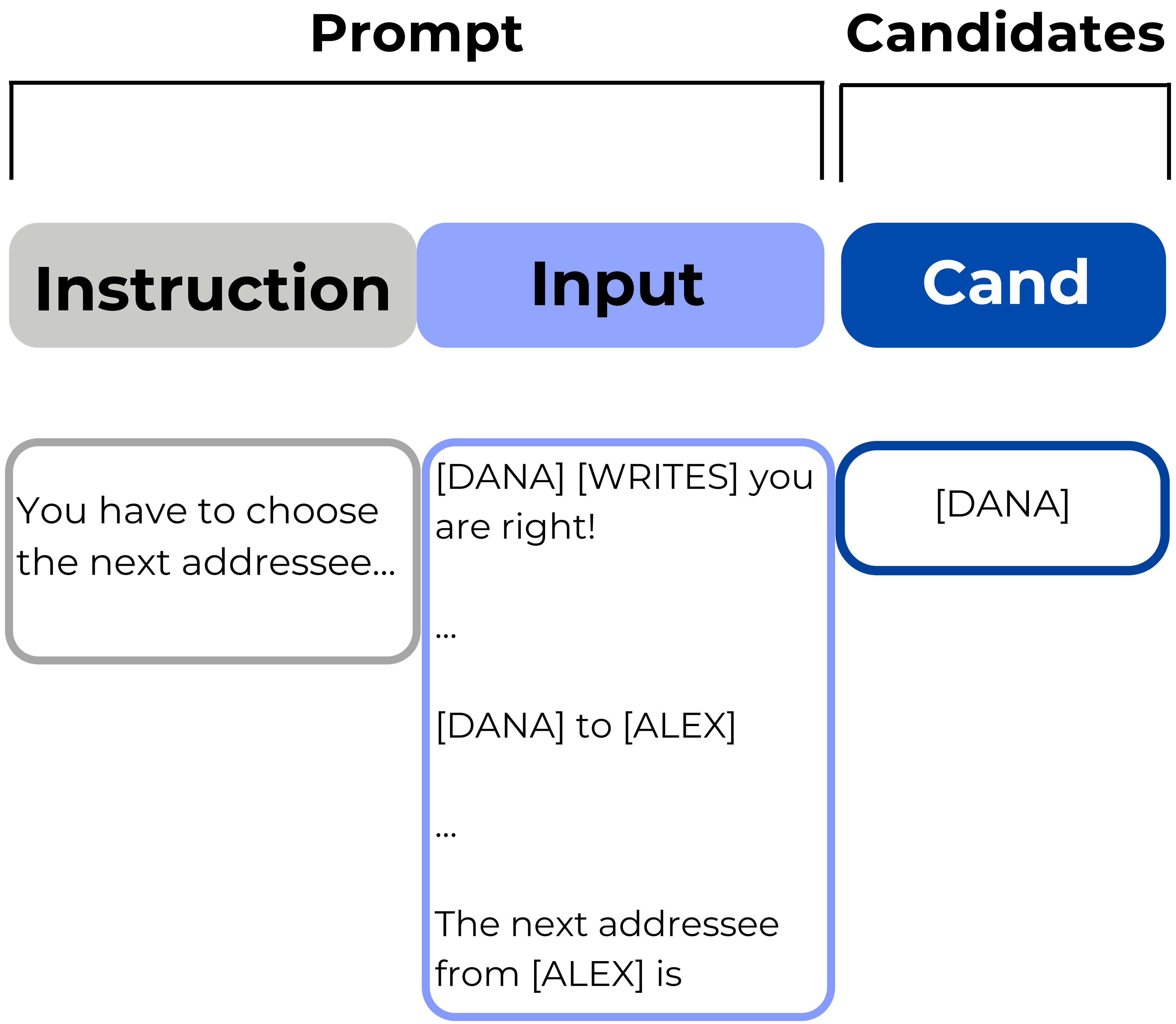
Response Selection

- Conversation Transcript
- Conversation Transcript | Interaction Transcript
- Summary | Interaction Transcript
- User Description | Interaction Transcript
- Summary | User Description | Interaction Transcript

Addressee Recognition

- Interaction Transcript
- Conversation Transcript
- Conversation Transcript | Interaction Transcript
- Summary | Interaction Transcript
- User Description | Interaction Transcript
- Summary | User Description | Interaction Transcript

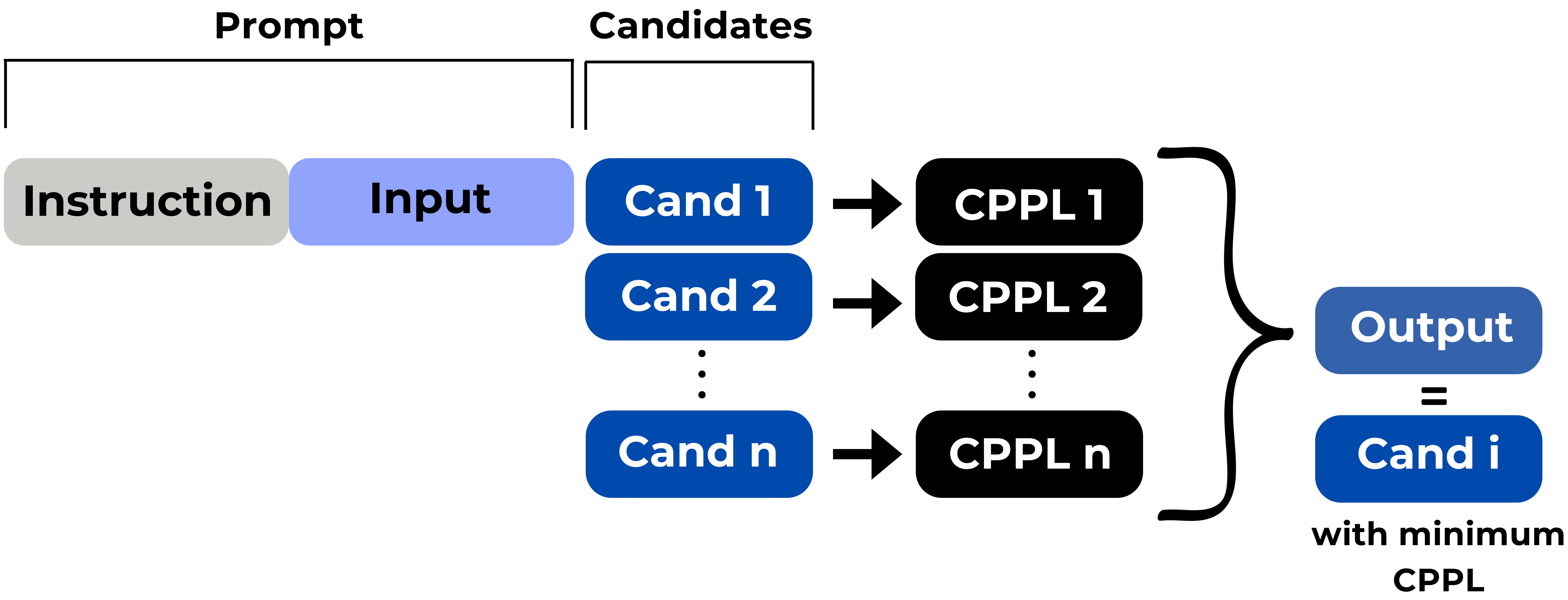
Classification



We compute the CPPL(Cand | Prompt)

The lower the CPPL, the better the candidate is!

Classification



Prompt Design

We tested different prompts in terms of Prompt Verbosity

Verbose

Each message is a string of text. Each message is associated with a speaker and an addressee. The speaker is the user who wrote the message. The addressee is the user to whom the message is directed. Each user can be the speaker of multiple messages and the addressee of multiple messages.

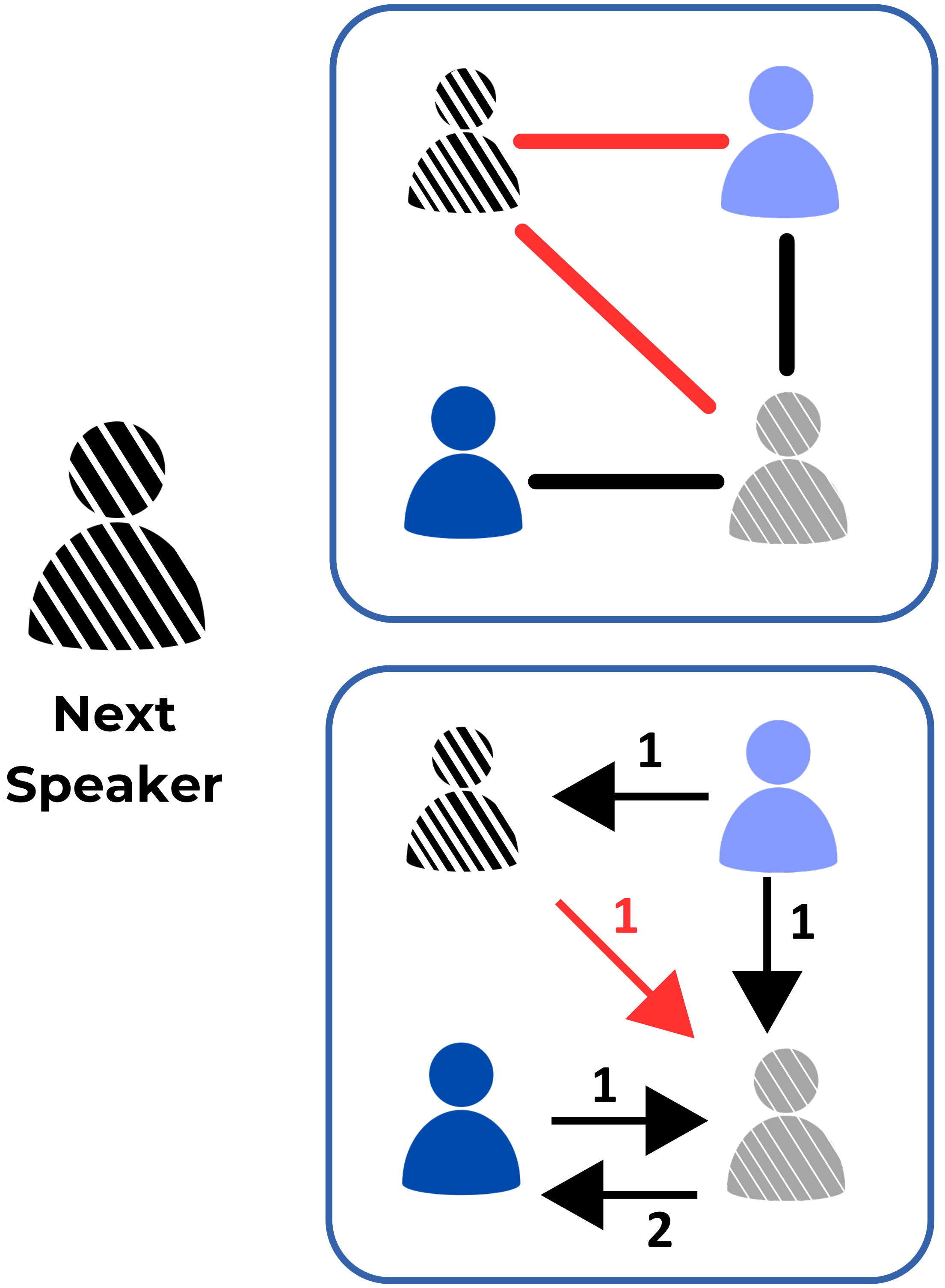
Medium

Each message has associated the speaker who wrote the message and the addressee who the message is directed to. Each speaker can write and be addressed by multiple messages.

Concise

Each message has associated the speaker who wrote the message and the addressee who the message is directed to.

Structural Analysis



Degree centrality → 2
of the "next speaker" node

Average Outgoing weight → 1
of the "next speaker" node

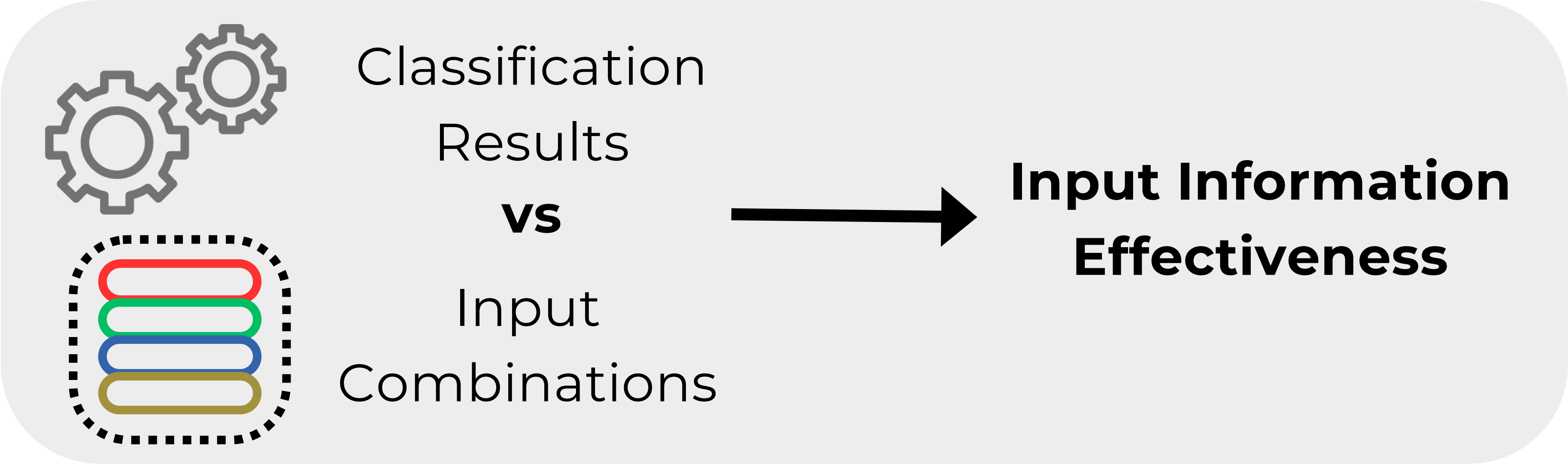
Evaluation

Classification Results **vs** Input Combinations → **Input Information Effectiveness**

Classification Results **vs** Levels of Prompt Verbosity → **Model Sensitivity to Prompt variation**

Classification Results **vs** Structural Analysis → **Performance across Structural Complexity**

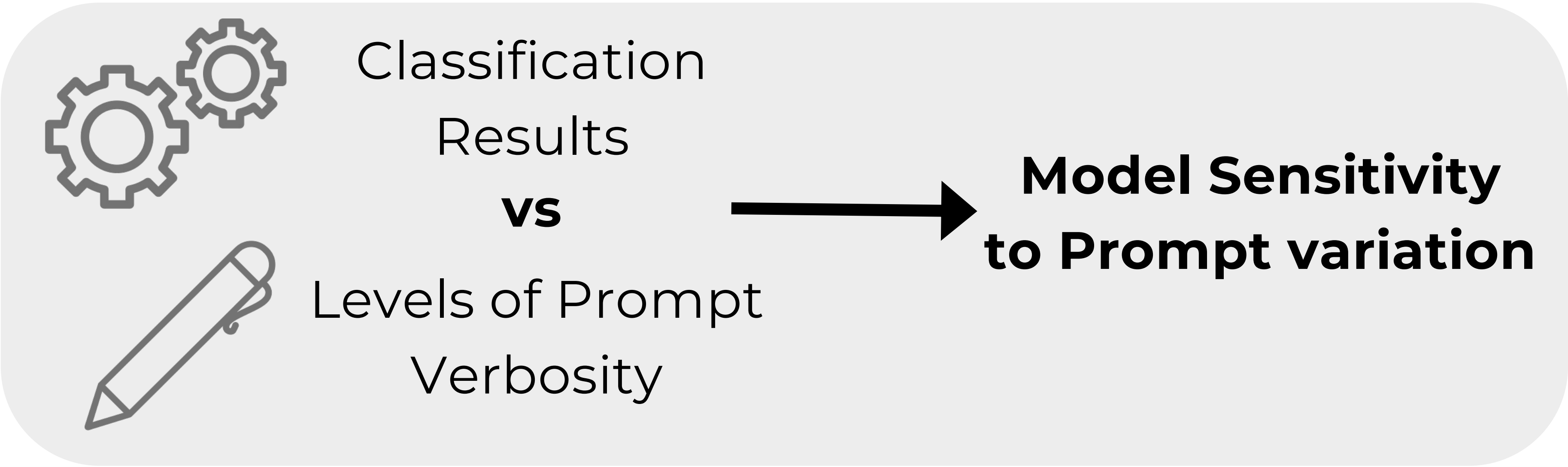
RQ1: Finding



Original information is **more effective** both in RS and AR.

Interaction transcripts are **crucial** for the **structural task** (AR), but **not** for the **linguistic task** (RS).

RQ2: Finding



The **structural task** (AR) shows **greater sensitivity** to prompts verbosity **compared** to the **linguistic task** (RS).

RQ3: Finding

Classification Results
vs
Structural Analysis → **Performance across Structural Complexity**

In the **structural task** (AR), performance **differences** across **input combinations** (always including Interaction Transcript) are due to **improved outcomes** on items of **low structural complexity.**

Take home messages

When dealing with Multi-Party Conversations, **do not rely only on macro accuracies** for assessing your model performance.

Only with a **diagnostic approach,**
you can really **understand your model capabilities.**

For **privacy reasons**, it is important to find a **tradeoff** between **privacy constraints and performance**.

Thanks for the Attention

PAPER

REPO

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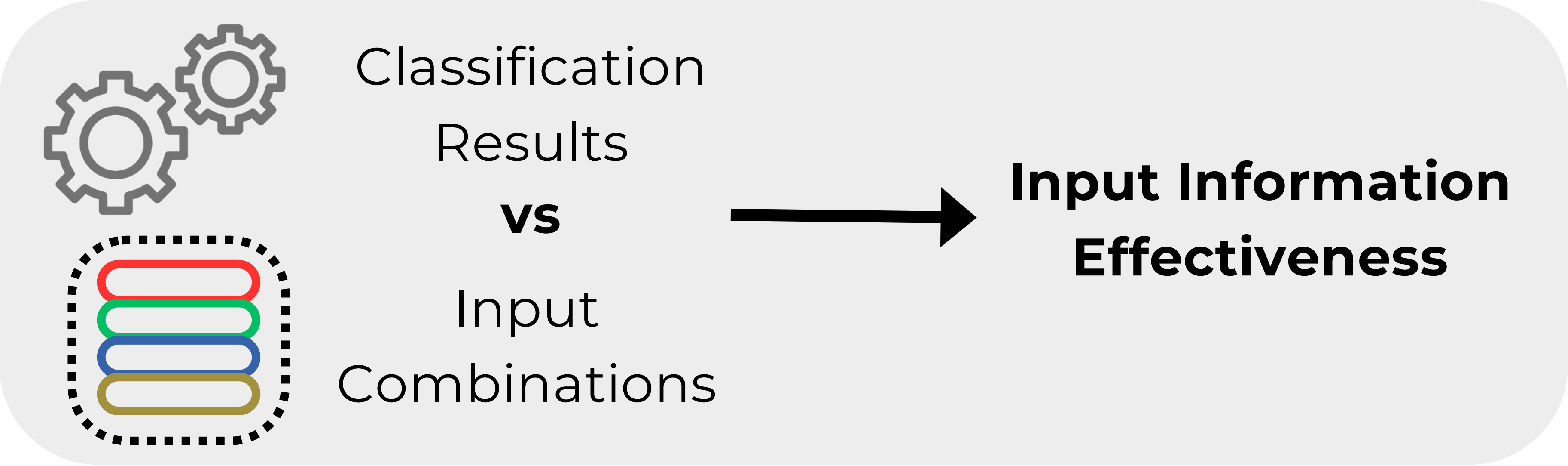
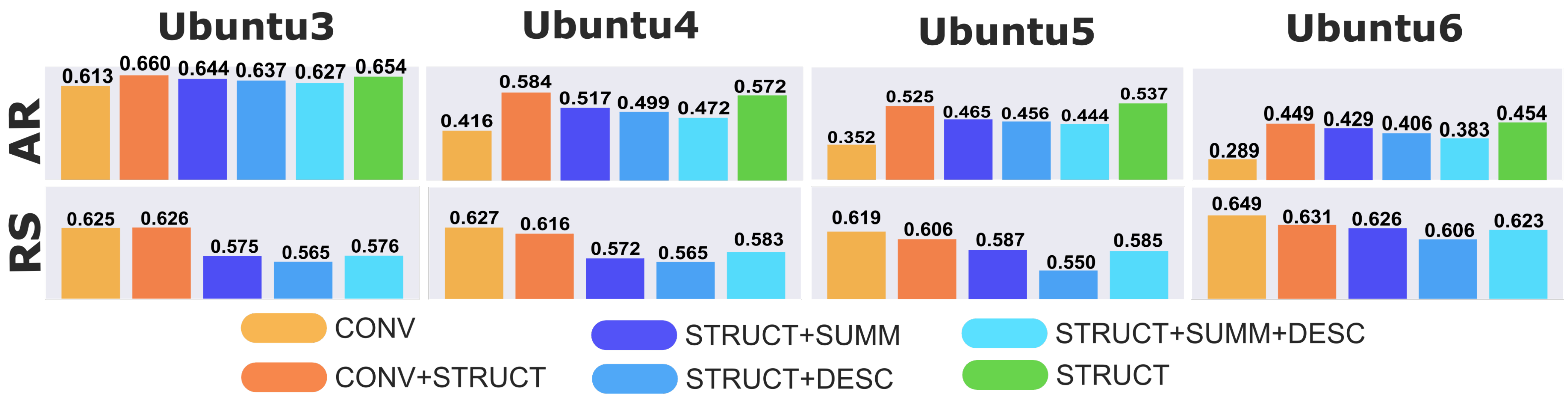
Additional Material

Evaluation


Classification Results
vs
Input Combinations
Input Information Effectiveness


Ubuntu3
Ubuntu4
Ubuntu5
Ubuntu6
AR
RS
0.613 0.660 0.644 0.637 0.627 0.654
0.416 0.584 0.517 0.499 0.472 0.572
0.352 0.525 0.465 0.456 0.444 0.537
0.289 0.449 0.429 0.406 0.383 0.454
0.625 0.626 0.575 0.565 0.576
0.627 0.616 0.572 0.565 0.583
0.619 0.606 0.587 0.550 0.585
0.649 0.631 0.626 0.606 0.623
CONV
CONV+STRUCT
STRUCT+SUMM
STRUCT+DESC
STRUCT+SUMM+DESC
STRUCT

Evaluation

Classification Results
vs
Levels of Prompt Verbosity

→ **Model Sensitivity to Prompt variation**

COMBINAT.	T	U.3	U.4	U.5	U.6
CONV	AR	2.7$^{\diamond}$	0.6	5.8$^{\diamond}$	2.0
	RS	0.8$^{\diamond}$	0.9$^{\diamond}$	0.8$^{\diamond}$	0.6
CONV + STRUCT	AR	7.1$^{\diamond}$	10.9$^{\diamond}$	4.5$^{\diamond}$	4.9$^{\diamond}$
	RS	0.4$^{\diamond}$	0.6	1.1*	0.8*
STRUCT+ SUMM	AR	2.5*	6.5$^{\diamond}$	3.6*	6.7*
	RS	0.8*	1.0	1.7	0.8$^{\diamond}$
STRUCT+ DESC	AR	2.7$^{\diamond}$	5.6$^{\diamond}$	4.8$^{\diamond}$	8.9$^{\diamond}$
	RS	2.1$^{\diamond}$	0.9*	1.3*	1.6
STRUCT+ SUMM+DESC	AR	1.5$^{\diamond}$	4.0$^{\diamond}$	3.2*	6.0$^{\diamond}$
	RS	0.2$^{\diamond}$	1.4	1.2	0.6$^{\diamond}$
STRUCT	AR	5.6$^{\diamond}$	8.3$^{\diamond}$	8.5$^{\diamond}$	6.3$^{\diamond}$

Table 1: Relative gap (%) between the best prompt result and the average, for each input combination and diagnostic dataset (UbuntuX is shortened as U.X), and for each task (i.e., AR and RS). We put a $\diamond$ when the best prompt is the verbose version, a $*$ when the medium version is the best and nothing when the best is the concise version.

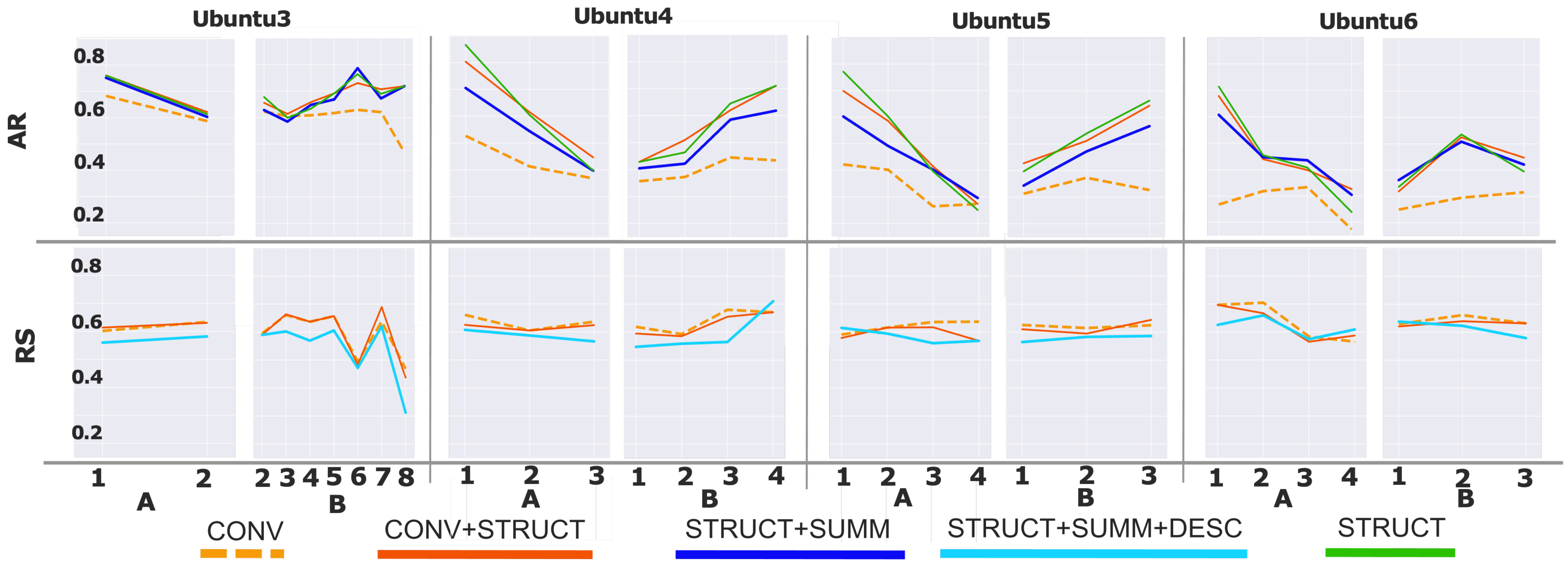
Evaluation

Classification Results

vs

Structural Analysis

→ **Performance across Structural Complexity**



A = degree centrality

B = average outgoing weight